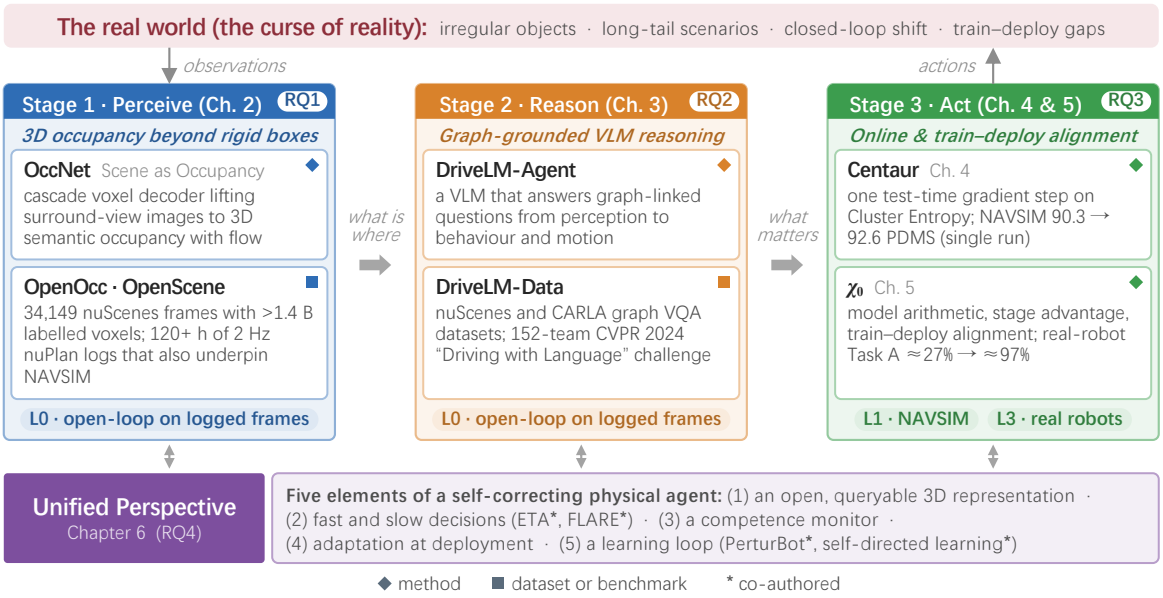

The real world (the curse of reality): irregular objects · long-tail scenarios · closed-loop shift · train–deploy gaps
observations
actions
Stage 1 · Perceive (Ch. 2) RQ1
3D occupancy beyond rigid boxes
OccNet Scene as Occupancy
cascade voxel decoder lifting surround-view images to 3D semantic occupancy with flow
OpenOcc · OpenScene
34,149 nuScenes frames with >1.4 B labelled voxels; 120+ h of 2 Hz nuPlan logs that also underpin NAVSIM
L0 · open-loop on logged frames
what is where
Stage 2 · Reason (Ch. 3) RQ2
Graph-grounded VLM reasoning
DriveLM-Agent
a VLM that answers graph-linked questions from perception to behaviour and motion
DriveLM-Data
nuScenes and CARLA graph VQA datasets; 152-team CVPR 2024 "Driving with Language" challenge
L0 · open-loop on logged frames
what matters
Stage 3 · Act (Ch. 4 & 5) RQ3
Online & train–deploy alignment
Centaur Ch. 4
one test-time gradient step on Cluster Entropy; NAVSIM 90.3 → 92.6 PDMS (single run)
χ0 Ch. 5
model arithmetic, stage advantage, train–deploy alignment; real-robot Task A ≈27% → ≈97%
L1 · NAVSIM
L3 · real robots
Unified Perspective
Chapter 6 (RQ4)
Five elements of a self-correcting physical agent: (1) an open, queryable 3D representation · (2) fast and slow decisions (ETA*, FLARE*) · (3) a competence monitor · (4) adaptation at deployment · (5) a learning loop (PerturBot*, self-directed learning*)
◆ method ■ dataset or benchmark * co-authored